Aim : Formulation And Calculation Of Ybus Matrix System Using Matlab Software

Input

```
clc
z12=input('input value of z12=');
z23=input('input value of z23=');
z24=input('input value of z24=');
z34=input('input value of z34=');
z13=input('input value of z13=');
y10=input('input value of y10=');
y20=input('input value of y20=');
y30=input('input value of y30=');
y40=input('input value of y40=');
y12=1/z12;
y23=1/z23;
y24=1/z24;
y34=1/z34;
y13=1/z13;
y14=0;
y21=y12;
y32=y23;
y42=y24;
y43=y34;
y31=y13;
y41=y14;
y11=y10+y12+y13+y14;
y22=y20+y21+y23+y24;
y33=y30+y31+y32+y34;
y44=y40+y41+y42+y43;
ybus=[y11 -y12 -y13 y14,
    -y21 y22 -y23 -y24
    -y31 -y32 y33 -y34,
    y41 -y42 -y43 y44];
```

Output

input value of z12=0.2+0.8j
input value of z23=0.3+0.9j
input value of z24=0.25+1j
input value of z34=0.2+0.8j
input value of z13=0.1+0.4j
input value of y10=0.03j
input value of y20=0.09j
input value of y30=0.06j
input value of y40=0.06j

y10	0.0000 + 0.0300i
y11	0.8824 - 3.4994i
y12	0.2941 - 1.1765i
y13	0.5882 - 2.3529i
y14	0
y20	0.0000 + 0.0900i
y21	0.2941 - 1.1765i
y22	0.8627 - 3.0276i
y23	0.3333 - 1.0000i
y24	0.2353 - 0.9412i
y30	0.0000 + 0.0600i
y31	0.5882 - 2.3529i
y32	0.3333 - 1.0000i
y33	1.2157 - 4.4694i
y34	0.2941 - 1.1765i
y40	0.0000 + 0.0600i
y41	0
y42	0.2353 - 0.9412i
y43	0.2941 - 1.1765i
y44	0.5294 - 2.0576i
ybus	*4x4 complex double*
z12	0.2000 + 0.8000i
z13	0.1000 + 0.4000i
z23	0.3000 + 0.9000i
z24	0.2500 + 1.0000i
z34	0.2000 + 0.8000i